

Lecture 5

Writing Progress Updates

Assignment FAQs

- Will all the assignments and the presentation be on the same analysis?
 - Yes
- Do we need to do the analysis or can we make up results
 - You have to do the analysis
- What is the presentation
 - You will present your results to one of three audiences (chosen by me)
 - Your peers and team leads in a data science team
 - Business stakeholders with an interest in their project
 - Team leads / managers at a different company
 - You will ask questions of the presenting group as that audience!

What is a Progress Update

A progress update is a concise report that explains the **status**, **accomplishments**, **challenges**, and **next steps** of a project

The Rhetorical Situation

Consider progress update you are delivering to your project sponsor.
What do we know about the rhetorical situation for our response?

As a reminder, we are looking for exigence, audience, purpose,
constraints, and context

The Rhetorical Situation

Exigence	
Audience	
Purpose	
Constraints	
Context	

Components of a Progress Update

- Introduction
 - Brief overview of the project
- Progress Summary
 - Completed tasks since the last update
 - Current status vs. planned objectives
- Recent Findings
 - New results of findings since the last status update
 - New questions or uncertainties that you want to address
- Next Steps
 - Upcoming and ongoing tasks
 - Requests for support

Progress Summary

- A progress summary serves two purposes
 - State what activities you have done and what activities you will do next
 - Provide an overview of the overall progress in the project
- DO include
 - Completed and planned tasks
 - Deliverables
 - Work-in-progress items
- DON'T include
 - Results
 - Technical details
 - Blockers

Progress Summary – Example 1

Project Updates



Recently Completed

- Setting up and accessing development environment
- Initial evaluation of key features of each document type
- Initial models of different classification approaches
- Ideation for combining different models

Currently In Progress

- Testing and evaluation of classification approaches
- Evaluating options for optical character recognition on scanned documents
- Testing options for combining models
- Planning for full-scale implementation
 - Handling non-PDF doc types
 - Defining work items
 - Estimating compute requirements

Progress Summary – Example 2

Milestone	Start Date	Completion Date	Progress
Kickoff	Date	Date	Completed
Business Understanding & Problem Definition	Date	Date	Completed
Exploratory Data Analysis	Date	Date	Completed
Create components for XXXXXX	Date	Date	Completed
Create components for critical and novel aspects XXXXX	Date	Date	In Progress
Delivery and acceptance of solution design	Date	Date	In Progress
Build, train, test, and evaluate refined model	Date	Date	On Track

Recent Findings

- Recent findings will usually be the most substantial part of your report
- The level of detail to include depends on your primary audience
 - Managers / line-of-business leaders will want to know how your findings will impact them
 - Data science leaders will want to know what your pipeline is and how you are evaluating
- Progress updates aren't meant to be exhaustive – stick to the highlights

Recent Findings – Example 1

Classification by Key Phrases



Document Type	Precision	Recall	Support
	100%	32%	71
	100%	44%	43
	100%	85%	100
	100%	71%	100
	100%	39%	100
	100%	69%	94
	91%	84%	100
	100%	27%	15
	96%	95%	76
WEIGHTED AVG	97%	67%	699

Precision: Of the samples where the model predicted this class, how many are correct?

Recall: Of all the samples of this class, how many were correctly identified?

Support: The number of samples evaluated

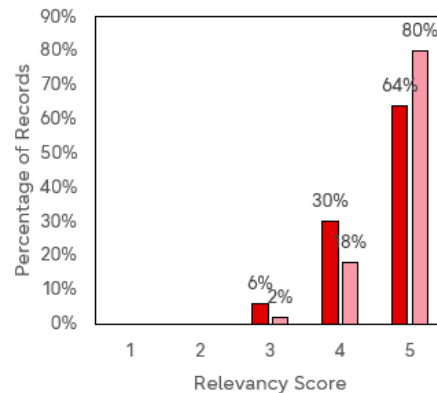
- Very high precision, the model is not misclassifying documents
- 462 of 699 samples were correctly identified
 - 8 identified as
 - 1 & 2 identified as
 - 226 documents not classified
- Most failures are because the text was not extracted properly

Recent Findings – Example 2

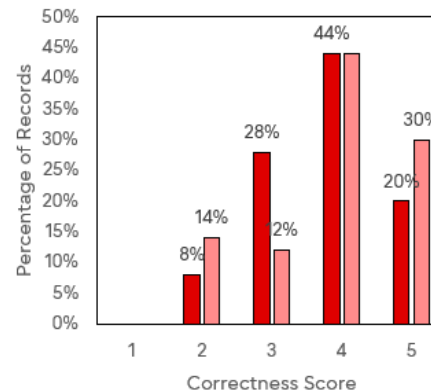
Evaluation Results



>95% of answers are relevant to the query



65-75% of answers are mostly or entirely correct



30% of answers are not certain

	Not Faithful	Faithful
Not Certain	22%	12%
Certain	18%	48%

	Not Faithful	Faithful
Not Certain	26%	10%
Certain	42%	22%

42% of  answers are certain but not supported

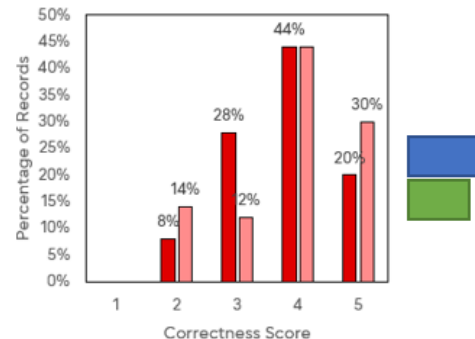
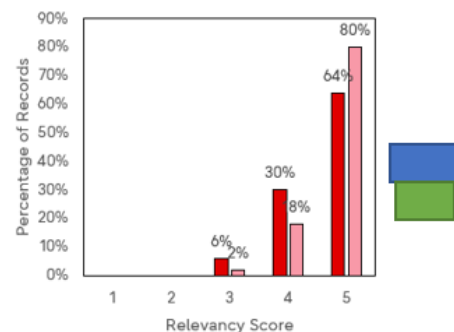
- **Relevancy:** Does the response answer the question?
- **Faithfulness:** Does the response contain information that is not present in the supporting context?
- **Correctness:** Does the response match the question and the reference answer?

Recent Findings in Written Updates

We have tested the current model on two test datasets, blue and green, with 200 items each. We evaluated them with three key metrics:

- **Relevancy:** Does the response answer the question?
- **Faithfulness:** Does the response contain information that is not present in the supporting context?
- **Correctness:** Does the response match the question and the reference answer?

Results are given in the figures below:



	Not Faithful	Faithful
Not Certain	22%	12%
Certain	18%	48%
	Not Faithful	Faithful
Not Certain	26%	10%
Certain	42%	22%

These results show that the model generates relevant content in most cases. <More explanation>

Next Steps

- List any tasks you are currently working on, or will work on before the next update
- Whenever you can, link your next steps to your recent findings
 - *e.g.* we will evaluate different system prompts and agentic methods to reduce the likelihood of a certain-but-not-faithful answer
- Highlight any blockers or requirements from the sponsor
 - *e.g.* We will expand our testing to purple and red test sets when we receive them

Best Practices

- Use bullet points, tables, and figures for clarity
- Avoid technical details
 - If part of your audience wants this detail, consider adding it in an appendix
 - Metrics are good, but they need to be explained
- Focus on process and impact

Common Mistakes

- Including too many details
- Leaving out things that didn't work
 - There are no negative results in data science!
- Not explaining why you did things

Assignment

For your assignment, you will write a status update on the project you proposed last week. This should be a written report (not Powerpoint) written for two audiences:

- **Primary:** The sponsor of your project, who is familiar with the topic but not with data science practices and terminology
- **Secondary:** Other data scientists within your team who are technically oriented but unfamiliar with your work

Reports should be brief and cover the key activities you have done in the past week and will do next week.